

Wage Effects of Local Industry-Level Minimum Wages:

Evidence from India

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Motivation

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Research Field Overview

- Minimum wages aim to **protect low earners**, but their effectiveness in dualistic labor markets is debated (Card and Krueger, 1994, Belser and Rani, 2011, Autor et al., 2016, Neumark and Corella, 2021, Engbom and Moser, 2022).
- **Multiple wage minima**: common in developing economies and are often set at state or sector level (Eyraud and Saget, 2005), raising key questions:
 1. How do minimum wages affect the lowest-earning workers when the definition of "lowest earner" varies across states and sectors?
 2. Do more specialized (state- or sector-level) minimum wages make it more likely that minimum wage policies positively impact informal workers (*the "Lighthouse Effect"*) by better adapting to local labor markets needs?

Research Context: India's flexible minimum wage system, set locally by sector under the 1948 Act, is designed to cover both formal and informal workers (Mansoor and O'Neill, 2021)—providing a unique setting to study these questions.

Minimum Wage Setting In India

Wage-Setting Structure:

- India is a federal country; each state's labor bureau sets and enforces its own minimum wages.
- Minimum wages are set at the industry level, resulting in multiple minimum wages per state.
- The system was designed with the goal of raising wages for formal *and informal* workers (Khurana et al., 2023, Mansoor and O'Neill, 2021).

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Features of the Minimum Wage

- All states set a minimum **daily** wage rate (based on an 8-hour workday).
- States are required to revise their minimum wages every 5 years.
- Sector-state minimum wages are declared to the Central Labor Bureau annually
→ The archive of reports are the data source for this project.

Fig. A1: Manufacturing MW by State

Fig. A2: Agriculture MW by State

Empirical Approach

Primary Identification Concerns:

- Reverse causality: sectors and states with higher average wages → higher minimum wages are set → Overestimation of the effect.
- Concern of omitted variable bias: factors influencing the selection of a minimum wage are not (fully) captured by my data.
- Concern of spillovers → workers moving to sectors (or states) to benefit from higher minimum wages. → Robustness: Total Effect

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Robustness: Check results with alternative fixed effects, new DiD estimators (Callaway and Sant'Anna, 2021, Baker et al., 2022), and border-pair DiD (Dube et al., 2010).

Contribution to the Literature

Preview of Findings

- Minimum wage hikes raise formal wages but lower informal wages, resulting in a null average effect.
- Hours worked adjust more than employment: employers may substitute informal for formal labor after minimum wage increases.

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Labor Market Dualism and Regulation

- Extends prior work on labor regulation in dualistic markets (Magruder, 2013, Almeida and Carneiro, 2012, Ulyssea, 2010, Ham, 2018) by leveraging India's unique state-sector wage setting and informality data.

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Wage Impacts of Multiple Minimum Wages

- Fills gaps on dual state-sector minimum wage variation, extending beyond studies of national or single-level regulation (Khurana et al., 2023, Engbom and Moser, 2022, Soundararajan, 2019, Leckcivilize, 2015, Alaniz et al., 2011).

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Labor Market Characteristics and Minimum Wages

- Builds on research of how local labor market characteristics mediate minimum wage effects (Azar et al., 2022, Arnold, 2019, Rinz, 2022, Badaoui and Walsh, 2022) by exploring differences between formal and informal labor markets.

Data Collection

Report on the Working of the Minimum Wage Act, 1948

The Indian Labor Bureau published annual reports between 2004 and 2019 listing state- and sector-level minimum wage rates, but no public database exists.

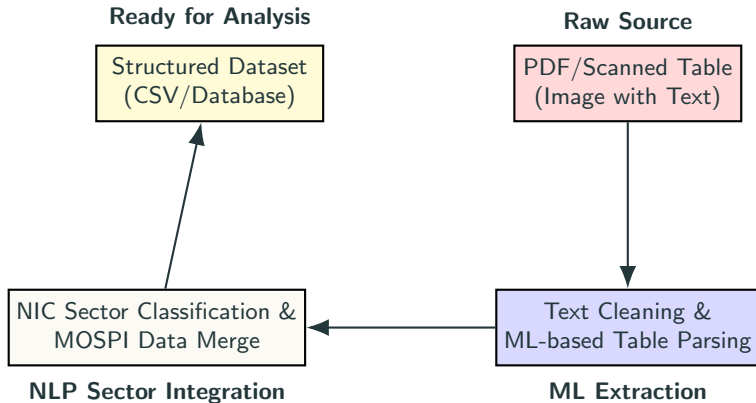
TABLE 5

TABLE 5
COMPARATIVE MINIMUM WAGE RATES PREVAILING IN SCHEDULED
EMPLOYMENTS IN CENTRAL SPHERE/STATES/UNION TERRITORIES
AS ON 31.12.2009

SL. NO.	NAME OF SCHEDULED EMPLOYMENT	CENTRAL SPHERE/ STATES/ UNION TERRITORIES	MINIMUM WAGE RATE (Rs.) PER DAY
1	Agriculture	1 A & N Islands	A-156.00 N-167.00
		2 Arunachal Pradesh	80.00
		3 Bihar	99.00
		4 C.L.C(Central)	130.00
		5 Chandigarh	157.36
		6 Delhi	152.00
		7 Haryana	151.00
		8 Himachal Pradesh	110.00
		9 Lakshadweep	141.85
		10 Manipur	81.40
		11 Meghalaya	100.00
		12 Mizoram	132.00
		13 Punjab	136.79
		14 Rajasthan	100.00
		15 Uttarakhand	106.25

Solution: Extract data from reports using a text processing algorithm and use NLP to match legal sector definitions with industrial classifications.

From PDF Tables to Usable Data: Extraction Pipeline



Pipeline: PDF → OCR → ML Table Extraction → NLP Sector Matching → Structured Dataset

Evolution of Minimum Wages Over Time

Ministry of Statistics and Program Implementation (MOSPI)

Survey Data:

Labor Force Surveys:

- **Unemployment/Employment Survey:** 2004 - 2012, **Periodic Labor Force Survey:** 2017 - 2019.
- Annual, repeated cross-section of individuals (UE/E), within-year panel with quarterly survey frequency (PLFS)
- **Variables:** labor force status, wages and working time for reference week, industry of employment, eligibility for benefits, type of contract, education/training characteristics. **LFS: Summary Statistics**

Enterprise Surveys:

- **Annual Survey of Industries (ASI)** 2004 - 2019, **Unorganised Enterprise Survey (UES):** 2005, 2010, 2015 → Repeated cross-sections.
- Covers non-agriculture in all states/UTs, ASI covers formal firms UES covers informal firms.
- **Variables:** industry, cost share/amount of inputs used, share of foreign capital, local share of employment **ASI/UES: HHI Calculation**.

Identification Strategy

Identification Strategy

Methodology: Staggered, two-way fixed effect (TWFE) strategy/DID with a continuous treatment and differential treatment timing.

- Following Angrist and Imbens (1995), Callaway et al. (2024), $\hat{\beta}$ is the average causal response (ACR) of Y to the minimum wage.
- The treatment effect is determined by (i) being exposed to a minimum wage increase and (ii) the magnitude of the increase (“dosage”).

Baseline Models:

Baseline Model:

$$Y_{i(ks)t} = \alpha + \beta \log(\text{Minimum Wage})_{k(s)t} + X_{i(ks)}\beta + \gamma_t + \delta_{ks} + \epsilon_{ikst} \quad (1)$$

- $Y_{i(ks)t}$ is the **log daily wage** of individual i in sector k in state s for time t .
- $X_{i(ks)}\beta$ are time-invariant worker controls including **age, gender, and education**.
- γ_t is a year fixed effect and δ_{ks} is a sector-state fixed effect.
 - The coefficient of interest is identified on **within sector-state unit variation** in the minimum wage and daily wages over time.

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Interaction Model:

$$Y_{i(ks)t} = \alpha + \beta \log(\text{Minimum Wage})_{kst} + \eta Z_{i(kst)} + \tau \log(\text{Minimum Wage})_{kst} \times Z_{i(kst)} + X_{i(ks)}\beta + \gamma_t + \delta_{ks} + \epsilon_{ikst} \quad (2)$$

Where the treatment is interacted with an indicator variable for **an individual characteristic (i.e., formal/informal, wage decile, gender/caste, education category)**.

Balance Test: MW Exposure

Baseline Results

Baseline Results

1. Results from Equation (1) show a null average effect of minimum wage laws on workers wages.
2. Null result suggests **2 potential hypotheses**:
 - 2.1 **H1**: Minimum wage laws effect formal and informal workers differently, driving the null average result (Lindbeck and Snower, 2001)
 - 2.2 **H2**: There is a redistributive effect and the negative effect at the top of the distribution conceals a positive effect at the bottom of the distribution (Engbom and Moser, 2022).

H1: Baseline Results and Informality

Table 1: Effect of Minimum Wage on Formal and Informal Workers: Various Wage Measures

	Baseline	Informality Interaction	Alternative Wage Measures	
	(1)	Daily Wage (2)	Weekly Wage (3)	Monthly Wage (4)
Log(MW)	0.0547 (0.150)	0.544*** (0.146)	0.112** (0.048)	0.148** (0.062)
Informal x Log(MW)		-1.486*** (0.158)	-0.499*** (0.086)	-0.646*** (0.109)
Informal		7.527*** (0.780)	3.103*** (0.475)	4.001*** (0.607)
Marginal Effects				
Formal Worker		0.544*** (0.146)	0.112** (0.048)	0.148** (0.062)
Informal Worker		-0.942*** (0.181)	-0.387*** (0.080)	-0.498*** (0.100)
Fixed Effects	Sector-State, Year			
Controls	Individual			
Observations	1,134,873			

Standard errors clustered at the sector-state level. Individual controls include age, gender, and level of education

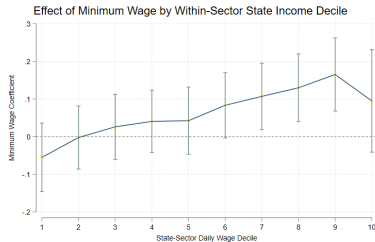
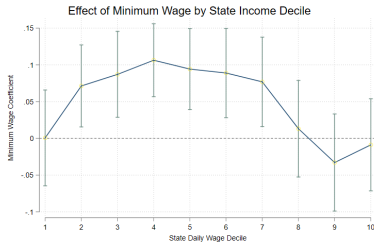
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Gender & Caste Heterogeneity

Education Level Heterogeneity

Migration Subsample

H2: Baseline Redistribution Effects



Note: Coefficients calculated using state-sector and year FE and using individual controls include age, sex, education level, and contract type. Confidence intervals represent cluster-robust standard errors at the sector-state level.

Distribution Relative MW Level

Distribution with Lagged MWs

Distribution with Self-Employment

Baseline: Informality vs. Redistribution

H1: Informality: significant and positive effect found for formal workers; and a significant negative effect found for informal workers → contrary to the Lighthouse Effect predicted by [Magruder \(2013\)](#), minimum wages may be insider-protecting.

- Analysis by demographic group (gender, caste) and triple interactions with informality confirm the main difference in minimum wage effects is still between formal and informal workers.

H2: Redistribution: results do not support a strong distributive effect, finding few differences across deciles. It may further support H1, showing a negative shock to income in the first decile.

- No evidence that lagged minimum wage increases lead to compression in the wage distribution over time.

Baseline Robustness Checks:

- Inflation adjustment CPI Adjustment
- Spillovers from other-sector MW increases Sector Spillovers
- State labor regulation rigidity Labor Regulations

Robustness Checks

Spillovers and Treatment Manipulation:

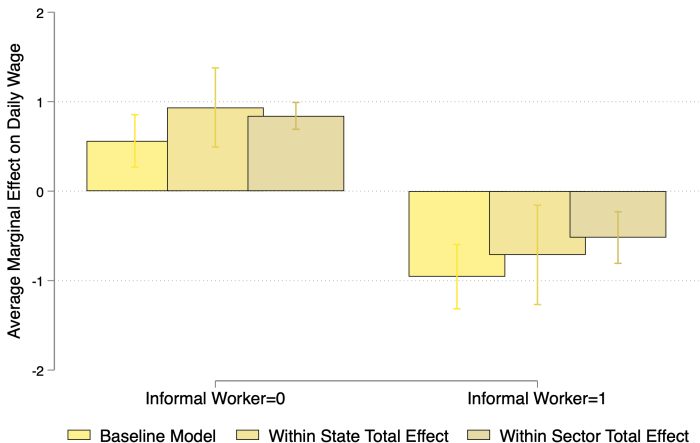
Using methodology from [Dustmann et al. \(2016\)](#): estimates the total effect of a minimum wage shock within-state or within-sector to account for state- or sector-switching by workers.

$$Y_{i(kp)t} = \alpha + \beta \log(\text{Minimum Wage})_{k(s)t} + X_{i(kp)}\beta + \gamma_{jt} + \epsilon_{ikpt} \quad (3)$$

Where fixed effects γ_{jt} are defined at the state-level (s) or sector-level (k) depending on the level of specification.

- Isolates within-state and within-sector variation to look at the **total effect** of a minimum wage shock for workers within an effected state or sector.
- Methodology introduced by [Dustmann et al. \(2016\)](#) to allow for the possibility that individual switch sectors within-state or states within-sector in pursuit of a higher minimum wage.

Results: Total Effect Approach of Dustmann et al. (2016)



Notes: The "Within State Total Effect" model includes state-year fixed effects, the "Within Sector Total Effect" model includes sector-year fixed effects, and the "Baseline" model includes sector-state and year fixed effects. All models control for individual characteristics: gender, age, and education. Standard errors are clustered at the sector-state level.

Omitted Local Labor Market Characteristics

Using methodology from [Dube et al. \(2010\)](#): exploit variation in minimum wages across state borders using contiguous districts in an unaffected state as a control group.

$$Y_{i(kp)t} = \alpha + \beta \log(\text{Minimum Wage})_{k(s)t} + X_{i(kp)}\beta + \gamma_{pt} + \delta_{pk} + \epsilon_{ikpt} \quad (4)$$

- **Identifying Assumption:** contiguous districts are similar on unobservable characteristics.
- Where γ_{pt} and δ_{pk} are time-border pair fixed effects and pair-sector fixed effects capturing local sector characteristics and time variant characteristics of district pairs.
- The coefficient of interest is identified on within border-pair-sector variations in the minimum wage due to being in different states.

Figure: Map of Border Districts

Results: Border-District Specification (Dube et al., 2010)

Table 2: Alt. Treatment Group: Border District Pair Specification

	Log(Daily Wage)	
	(1)	(2)
Log(MW)	-0.101 (0.139)	0.313** (0.143)
Informal Worker × Log(MW)		-1.466*** (0.167)
N	352,778	352,778
FE	District Pair-Sector, District Pair-Year	
Controls	Individual	

Standard errors clustered by state-border district segment, individual controls contain age, sex, and education level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Mechanisms

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1. **Characteristics of Production:** No significant variation of effect by the average capital, trade, ICT, or labor-intensity of production within a sector-state unit for formal or informal firms. Results: Employer Characteristics

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3. **Effect on Employment:** while there is a null effect on employment for informal and formal workers, I find evidence of adjustment of hours worked per week. The direction of the effect diverges for formal and informal workers.
 - 3.1 Formal workers experience **a significant reduction in hours worked per week**, but informal workers experience **a significant increase**. Employment Effects
 - 3.2 Persistent negative effects on wages after adjusting for time worked suggests **formal workers are paid more but hired less**. Firms may offset costs by increasing hiring of cheaper informal labor hours and not increasing their pay.

Table 3: Disemployment and Working Hours Adjustment

	Pr(Unemployed = 1)		Hours Worked in Week		Log(Hourly Wages)	
	(1) Baseline	(2) Interaction	(3) Baseline	(4) Interaction	(5) Baseline	(6) Interaction
Log(MW)	-0.00006 (0.00369)	0.0003 (0.00384)	0.002 (0.0384)	-0.148*** (0.0528)	0.0379 (0.1000)	0.391*** (0.0985)
Informal Worker $\times$ Log(MW)		-0.00108 (0.00310)		0.498*** (0.0350)		-1.040*** (0.0963)
Marginal Effects						
Formal Worker		0.000303 (0.00384)		-0.148*** (0.0528)		0.391*** (0.0985)
Informal Worker		-0.000776 (0.00421)		0.350*** (0.0510)		-0.649*** (0.117)
N	1,134,873	1,134,873	1,134,873	1,382,335	1,134,873	1,382,335
Fixed Effects			Sector-State, Year			
Controls			Individual			

Standard errors clustered at the state-sector level. Individual controls include sex, age, educational level, and contract type for baseline models. In interaction models, individual controls include sex, age, and education levels.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Conclusion

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Key Findings:

- Increases in the minimum wage at the state-sector level are associated with **increases in wages for formal-sector workers, but decreases for informal-sector workers.**
- Potentially driven by the **employers exchanging formal worker-hours hired for informal worker-hours** thus avoiding a significant increase in labor costs.
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Future Research:

- Implementing new methods for using TWFE/Staggered DID with continuous treatment (**Callaway et al., 2024**).
- Improving employer-level data to improve picture of local labor market characteristics.
- Exploring hiring patterns in employer-level data to confirm employment findings from worker data.

Thank you for your attention.

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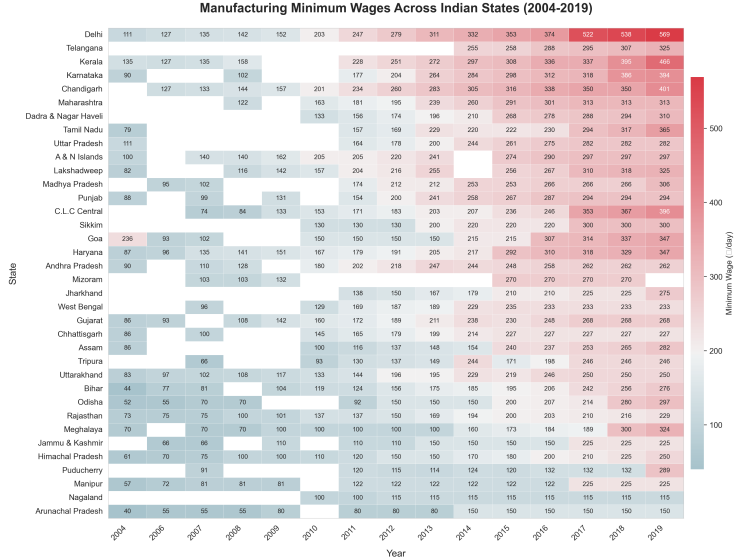
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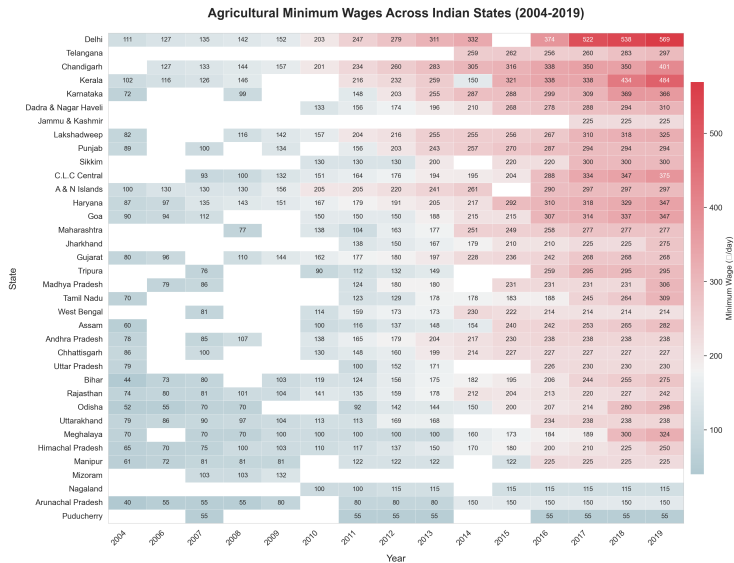
Appendix

Example: Manufacturing MW by State and Year



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Example 2: Agricultural MW by State and Year



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LFS Sample: Descriptive Statistics I/II

	Summary
N	2,963,285
Age	30.977 (18.249)
Gender	
1	1,768,868 (59.7%)
2	1,194,230 (40.3%)
3	187 (0.0%)
Formal Education (Years)	8.367 (5.248)
Secondary Employment	
0	2,708,835 (91.4%)
1	254,450 (8.6%)
Eligible for Paid Leave	
0	2,753,564 (92.9%)
1	209,721 (7.1%)
Eligible for Social Security	
0	2,824,639 (95.3%)
1	138,646 (4.7%)
Migrated in Past 5 Years	
1	14,449 (2.7%)
2	529,062 (97.3%)
Informal Worker	
0	2,150,568 (72.6%)
1	812,717 (27.4%)
Unemployed Worker	
0	2,905,018 (98.0%)
1	58,267 (2.0%)
Days Worked in Reference Week	3.752 (2.123)
Daily Wage	97.663 (293.572)

Enterprise Survey - Calculation of Labor Market Concentration

- The Herfindahl Hirshman Index (HHI) measures the size of firms in relation to other firms in the industry.
- Applicable to product and labor markets and measures the level of competition between firms in a market (for customers, workers, etc.).
- In this paper, it is the share of firms in the total employment for each sector-state unit.

$$\text{Employment Share of Firm} = \frac{\text{Employees of Firm } N}{\sum_{i=1}^N \text{Employees of Firm } i \text{ for Sector } K}$$

$$\text{Sectoral HHI} = \overline{\text{Employment Share of Firm}} \times 10,000$$

Following the approach of [Azar et. al \(2022\)](#), labor markets are characterized as follows:

- Low concentration: $\text{HHI} < 2500$
- Low moderate concentration: $2500 < \text{HHI} < 5000$
- High moderate concentration: $5000 < \text{HHI} < 7500$
- High concentration: $\text{HHI} > 7500$

Individual Characteristics: Balance by Exposure to MW

	= 1 if Works in a State-Sector with Minimum Wage		
	0	1	Total
N	1,360,539 (45.9%)	1,602,746 (54.1%)	2,963,285 (100.0%)
Age	30.691 (18.082)	31.227 (18.391)	30.977 (18.249)
Gender			
1	824,749 (60.6%)	944,119 (58.9%)	1,768,868 (59.7%)
2	535,726 (39.4%)	658,504 (41.1%)	1,194,230 (40.3%)
3	64 (0.0%)	123 (0.0%)	187 (0.0%)
Formal Education (Years)	8.582 (5.445)	8.236 (5.122)	8.367 (5.248)
Secondary Employment			
0	1,244,547 (91.5%)	1,464,288 (91.4%)	2,708,835 (91.4%)
1	115,992 (8.5%)	138,458 (8.6%)	254,450 (8.6%)
Eligible for Paid Leave			
0	1,252,862 (92.1%)	1,500,702 (93.6%)	2,753,564 (92.9%)
1	107,677 (7.9%)	102,044 (6.4%)	209,721 (7.1%)
Eligible for Social Security			
0	1,289,493 (94.8%)	1,535,146 (95.8%)	2,824,639 (95.3%)
1	71,046 (5.2%)	67,600 (4.2%)	138,646 (4.7%)
Migrated in Past 5 Years			
1	8,119 (2.6%)	6,330 (2.7%)	14,449 (2.7%)
2	301,558 (97.4%)	227,504 (97.3%)	529,062 (97.3%)
Informal Worker			
0	986,833 (72.5%)	1,163,735 (72.6%)	2,150,568 (72.6%)
1	373,706 (27.5%)	439,011 (27.4%)	812,717 (27.4%)
Unemployed Worker			
0	1,330,824 (97.8%)	1,574,194 (98.2%)	2,905,018 (98.0%)
1	29,715 (2.2%)	28,552 (1.8%)	58,267 (2.0%)
Days Worked in Reference Week	3.537 (2.054)	3.964 (2.168)	3.752 (2.123)

Robustness: Adjusting and Controlling for Inflation

Table 4: Minimum Wage Effect: Adjusting and Controlling for Inflation

	Baseline 2011-2019 Sample (1)	CPI as Control (2)	Log(Real Daily Wage) (3)
Log(MW)	-0.229 (0.191)	-0.136 (0.166)	
CPI		-0.0298*** (0.00782)	
Log(Real MW)			-0.00649 (0.0157)
N	423,627	501,972	509,306
FE		Sector-State, Year	
Controls		Individual	

Standard errors clustered at the state-sector level. Individual controls include age, sex, education level, and contract type.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Robustness: Other Sector Spillovers

Table 5: Spillover Effects on Wages for Other-Sector Minimum Wages

	(1)	(2)	(3)	(4)
Log (MW)	0.0000129 (0.142)		0.0291 (0.0488)	
MW Spillovers	-0.0108 (0.0247)	-0.00263 (0.0180)	-0.101 (0.0840)	-0.0227 (0.0391)
Observations	1,126,295	1,199,206	1,126,296	1,199,205
Fixed Effects		Sector-State, Year		
Controls		Individual		

Standard errors clustered at the state-sector level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Baseline Wage Effect Controlling for Number of MW in State

	(1)
Log(MW)	0.0496 (0.146)
Number of MW	0.00214 (0.00376)
Observations	1,134,873
Fixed Effects	Sector-State, Year

Standard errors clustered at the state-sector level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Robustness: Informality, Gender, and Caste

Table 7: Effect of Minimum Wage on Informal Workers: Gender and Caste Interactions

	(1) Gender x MW	(2) Caste x MW
Log(MW)	0.137 (0.161)	-0.035 (0.069)
Female Worker	0.312 (0.365)	
Female x Minimum Wage	-0.197*** (0.073)	
Scheduled Caste Member		0.165 (0.267)
Scheduled Caste x Log(MW)		0.037 (0.056)
Marginal Effects		
Female Workers	-0.060 (0.141)	
Male Workers	0.137 (0.161)	
Workers in Scheduled Caste		0.002 (0.084)
Workers Not in Scheduled Caste		-0.035 (0.069)
Fixed Effects	Sector-State, Year	
Controls	Individual	
Observations	1134835	889807

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Robustness: Educational Achievement

Table 8: Heterogeneity of Wage Effect by Education Level

	(1)
Dependent Variable: Log Daily Wage	
Log(MW) x	
Education Level = 1	-0.107 (0.138)
Education Level = 2	0.177 (0.152)
Education Level = 3	0.782*** (0.177)
Observations	1132948
Fixed Effects	Sector-State, Year
Controls	Individual

Standard errors clustered at state-sector level

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Robustness: Migration History Subsample

Table 9: Variation of Wage Effect by Migration Status: 2007-08 Subsample

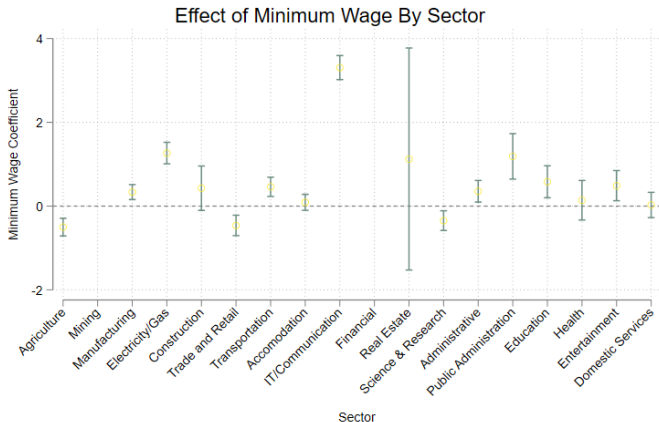
	Additional Control: Migrant	Interaction Model	Marginal Effect by Migration	Pr(Migrant == 1)
	(1)	(2)	(3)	(4)
Log(MW)	0.127 (0.0864)	0.743*** (0.192)		-0.00257 (0.00854)
Migrants	0 (.)	0 (.)	0.743*** (0.192)	
Non-Migrants	-0.291*** (0.0538)	2.555*** (0.788)	0.115 (0.0869)	
Non-Migrants × Log(MW)		-0.628*** (0.173)		
Observations	233,459	233,459	233,459	233,459
Fixed Effects		Sector-State, Year		
Controls		Individual		

Standard errors clustered at state-sector level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

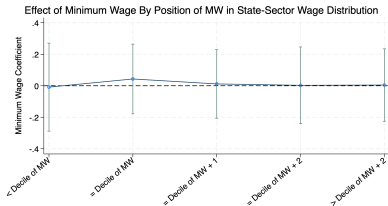
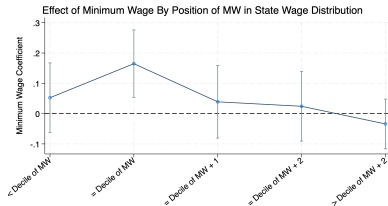
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Wage Effect Analysis: By Sector



Coefficients calculated using state-sector and year FE and using individual controls including age, sex, education level, and contract type. Confidence intervals represent cluster-robust standard errors at the sector-state level.

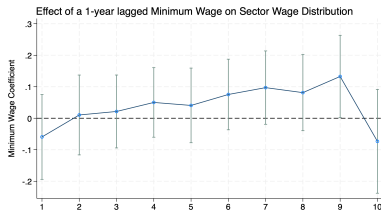
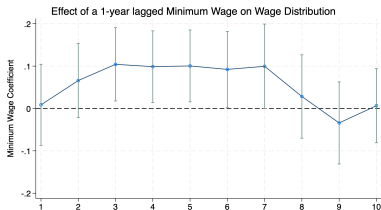
Wage Effect Near the Decile of the Minimum Wage



Note: Coefficients calculated using state-sector and year FE and using individual controls include age, sex, education level, and contract type. Confidence intervals represent cluster-robust standard errors at the sector-state level.

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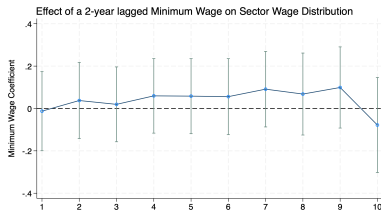
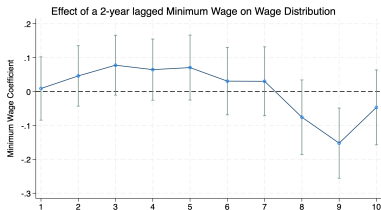
Redistributive Effect with a Lagged MW Treatment 1/2



Note: Coefficients calculated using state-sector and year FE and using individual controls include age, sex, education level, and contract type. Confidence intervals represent cluster-robust standard errors at the sector-state level.

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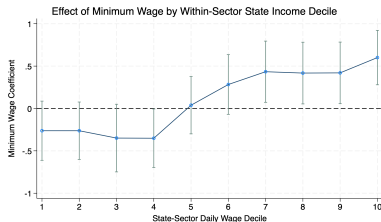
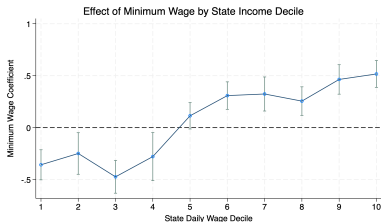
Redistributive Effect with a Lagged MW Treatment 2/2



Note: Coefficients calculated using state-sector and year FE and using individual controls include age, sex, education level, and contract type. Confidence intervals represent cluster-robust standard errors at the sector-state level.

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Redistributive Effect Including Self-Employed Workers

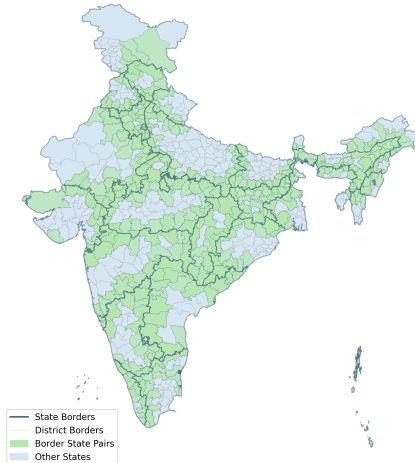


Note: Coefficients calculated using state-sector and year FE and using individual controls include age, sex, education level, and contract type. Confidence intervals represent cluster-robust standard errors at the sector-state level.

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District-Level Specification

Contiguous Border District Pairs - India



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Mechanisms: Average Formal Employer Characteristic

Table 10: Variation of Wage Effect by Employer Characteristics: Formal Sector

	Labor Market Concentration	Average Employer Factor Use			
	(1)	(2)	(3)	(4)	(5)
Formal Sector					
Log(MW)	-0.114 (0.133)	-0.454 (0.446)	-1.104** (0.440)	1.305 (16.08)	-0.654 (0.480)
× HHI < 2500	(.)				
× 2500 < HHI < 5000	-0.0344 (0.108)				
× 5000 < HHI < 7500	0.438** (0.177)				
× 7500 < HHI	0.0157 (0.286)				
× ICT Intensity		0.0386 (0.0344)			
× Physical Capital Intensity			0.0716*** (0.0264)		
× Trade Intensity				-1.158 (14.70)	
× Labor Intensity					0.121 (0.0818)
N	406057	225202	233115	55633	243055
Fixed Effects		Sector-State, Year			
Controls		Individual			

Standard errors clustered at state-sector level. Individual controls include sex, age, educational level, and contract type.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Mechanisms: Average Informal Employer Characteristic

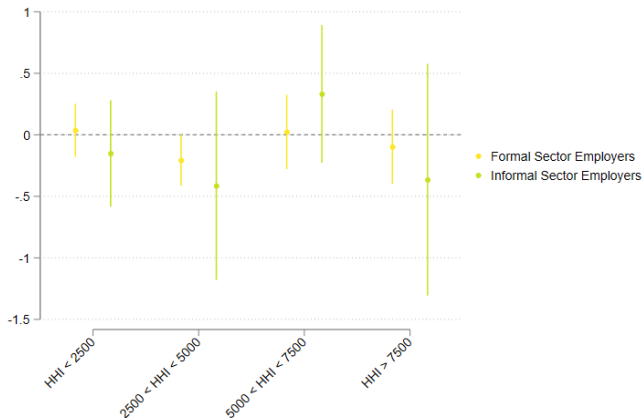
Table 11: Variation of Wage Effect by Employer Characteristics:
Informal Sector

	Labor Market Concentration	Average Employer Factor Use			
	(1)	(2)	(3)	(4)	(5)
Informal Sector					
Log(MW)	0.0531 (0.224)	-0.536* (0.254)	0.534 (0.702)	-0.889 (0.704)	-1.221*** (0.271)
× HHI < 2500	(.)				
× 2500 < HHI < 5000	-0.269 (0.270)				
× 5000 < HHI < 7500	0.512** (0.280)				
× 7500 < HHI	-0.0154 (0.245)				
× ICT Intensity		0.107*** (0.0265)			
× Physical Capital Intensity			-0.0356 (0.0556)		
× Trade Intensity				0.0698 (0.0608)	
× Labor Intensity					0.246** (0.0886)
Fixed Effects		Sector-State, Year			
Controls		Individual			

Standard errors clustered at state-sector level. Individual controls include sex, age, educational level, and contract type.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Mechanisms: Employer Labor Market Concentration



Notes: Follows the HHI thresholds for concentration defined in [Azar et al. \(2022\)](#). This model uses the baseline specification for fixed effects (sector-state, year). All models control for individual characteristics: gender, age, and education. Standard errors are clustered at the sector-state level.